UniGuide

Business Proposal

UMHackathon 2026, Final Round, Domain 1: AI Systems & Agentic Workflow Automation

Team: Breaking Bank, Team 108

Live deploy: uniguide-blush.vercel.app

Repository: github.com/ZenBen5173/uniguide

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1. Executive Summary

UniGuide is an AI co-pilot that ends the silent-fail problem in university administrative paperwork. Built on Z.AI's GLM as a hard architectural dependency, it walks Universiti Malaya students through five high-stakes procedures (scholarship applications, final-year projects, deferments, exam-result appeals, and postgraduate admissions) by reading the official Standard Operating Procedure (SOP) and the student's history, then asking for the next thing it needs, one step at a time.

Coordinators receive a pre-digested briefing per submission with extracted facts, SOP-cited red flags, an AI recommendation, and a confidence score. They preview AI-drafted decision letters with an automated hallucination check (a regex layer that verifies every name, matric, CGPA, year, faculty, and programme mention against the student's record before send), then approve, reject, or request more info, with a five-minute undo (a window during which the coordinator can take the decision back before it commits permanently). Students see decisions and letters in their portal in real time, and every letter is rendered as a downloadable, printable PDF on UM letterhead.

Beyond the hackathon, UniGuide is a B2B-to-University SaaS opportunity. Malaysia has 20 public universities serving roughly 1.2 million students; ASEAN serves more than 12 million. Each university spends substantial coordinator time on procedural rework that AI grounding can eliminate. We propose a pilot with UM that scales to other Malaysian public universities, then to private institutions, then to the broader ASEAN region, with ILMU (the Malaysian-sovereign LLM by YTL AI Labs and UM) as a structural advantage for government-sector procurement.

2. Problem Statement

UM students lose semesters to silent-fail paperwork. Scholarship applications miss income-tier rules buried in paragraph six of an SOP PDF. Final-year-project registration stalls for weeks because supervisor confirmation arrives in the wrong format. Deferment requests bounce between three offices because the eligibility ladder is non-obvious. The cost is borne overwhelmingly by students from families that cannot afford a wasted year.

The system is also costly for the institution. Coordinators triage hundreds of submissions per intake, repeating the same eligibility checks by eye. Response deadlines (Service-Level Agreements, or SLAs, the response-time commitment per case) slip because the inbox is sorted oldest-first rather than risk-weighted. Decision letters are hand-written from templates, so wording inconsistencies slip past review. Audit traceability is patchy: there is no single record of how a particular decision was reached.

Existing solutions do not close the gap. Generic chatbots (ChatGPT, Claude, Gemini) hallucinate UM-specific policy and have no integration with student records. Prior in-house chatbot pilots have been form-filling layers, not planners; they cannot adapt to a student whose situation diverges from the standard case (the simple path with no unusual circumstances).

3. Solution

UniGuide is an end-to-end loop, not a chatbot. The full student to AI to coordinator to letter cycle works on the live site for all five MVP procedures, with the AI doing four load-bearing things on the application's behalf:

- **Planning the next question.** GLM reads the SOP excerpts and the student's history at every turn and emits the next step with citations to the SOP sections it relied on. There is no static template and no hard-coded branches.
- **Document understanding.** Uploaded PDFs (income proofs, transcripts, supporting letters) are text-extracted and structured-parsed inline so the AI doesn't re-ask for facts it can read directly.
- **Coordinator briefing.** GLM digests every submission into facts, SOP-cited flags, a recommendation and a confidence score so the coordinator sees in seconds what would otherwise take ten minutes of manual review.
- **Letter generation with hallucination check.** GLM fills the procedure's letter template; a regex layer verifies every name, matric, CGPA, year, faculty, and programme mention against the student's record before send. The AI does the heavier writing work; the coordinator reads the draft, edits if needed, and signs off before anything reaches the student.

Two further co-pilot surfaces extend the product: a coordinator-side natural-language Revise channel that lets staff iterate on AI output by instruction (scoped to one application), and an always-on student AI chat that answers SOP questions while the student fills the form and one-click-escalates to a coordinator when the situation requires human discretion.

4. Market Analysis

Three concentric markets, each independently viable.

4.1 Universiti Malaya (Starting Market)

UM enrols approximately 23,000 undergraduates and 19,000 postgraduates (FY24 enrolment), totalling around 33,000 active students once part-time and continuing-education enrolments are included. The Hal Ehwal Pelajar (HEP) office, faculty academic offices, and the Institute of Graduate Studies (IGS) collectively process tens of thousands of administrative procedures per year. Coordinator headcount is thin relative to volume, and the institution has visible appetite for AI augmentation through partnerships with YTL AI Labs (the ILMU programme) and with Z.AI.

4.2 Malaysian Public Universities (Phase 2)

Twenty public universities across Malaysia serve roughly 1.2 million students. UPM, USM, UKM, UTM, UM, UIA, and UPSI are the largest by enrolment. Procedures are broadly similar across institutions because they share regulatory parents (MOHE, JPA, EMGS for international students). A pattern that works for UM should generalise with low per-institution customisation cost, primarily SOP uploads and letter-template edits.

4.3 ASEAN Universities (Phase 3)

Indonesia, Vietnam, Thailand, the Philippines, and Singapore collectively host more than 12 million tertiary students. The procedural pattern is similar; the regulatory parent differs. A multi-tenant rebuild is required for this phase (out of MVP scope; see roadmap). The Z.AI GLM dependency localises to languages used in each market; ILMU is a regional credibility signal for Malaysian government procurement specifically.

5. Competitor Analysis

Alternative	What it does	Why UniGuide wins
UM existing portals (MAYA, SiswaMail, SPeCTRUM, AskUM)	Systems of record. Forms, status pages, and document storage.	UniGuide complements rather than replaces them. We sit on top, plan the journey, and call into them as integrations later.
Generic LLM chatbots (ChatGPT, Claude, Gemini)	Open-domain question answering.	Hallucinate UM-specific policy. No integration with student records. No SOP-grounded citations. No coordinator briefing or letter generation. No audit trail.
In-house form-filling chatbots (prior pilots)	Static decision trees.	Cannot adapt when a student's situation is unusual. Cannot read SOP updates without engineering effort. UniGuide replans dynamically on every step.
Custom-built integrations	Custom-made workflows for each procedure.	Cost-per-procedure scales linearly with the number of procedures. UniGuide's planner is the same code for every procedure; only the SOP and letter templates change.
Doing it the old way (paper forms, email, in-person visits)	Manual rework, repeated requests, lost semesters.	UniGuide reduces coordinator time per case from ~10 minutes to ~2 minutes, and student time-to-decision from weeks to hours.

6. Value Proposition

- **For students:** an AI helper that already knows the SOP, your transcript, your income tier, and the language you are most comfortable in. No more stalled applications. No more silent rejections.
- **For coordinators:** a pre-digested triage queue. SOP-cited flags. AI-drafted letters with hallucination-proofing. Five-minute undo. Talk to the AI in natural language to revise drafts. Bulk-approve clean cases in one click.
- **For administrators:** complete audit trail of every AI decision (per-call input, output, latency, tokens, confidence). Live token-cost dashboard. Pluggable AI provider routing: Z.AI GLM by default, ILMU for Bahasa-heavy work.
- **For UM as institution:** higher decision throughput, lower coordinator burnout, defensible auditability for compliance, and a reference site that proves Malaysia can build AI-native public services on the Malaysian-sovereign LLM stack (ILMU).

7. Business Model

7.1 Pricing (B2B SaaS, per-student-seat)

The core model is per-active-student-per-year, billed annually to the institution. Three tiers:

Tier	Procedures included	Price (RM / student / year)	Best for
Pilot (Year 1, UM)	All 5 MVP procedures	Co-funded with UM Centre of Innovation; nominal per-seat	Reference site, case study, validation
Growth	All MVP plus 5 additional procedures	RM 8 / student / year	Mid-size public universities (UTM, USM)
Enterprise	Unlimited procedures plus custom integrations	RM 15 / student / year plus setup fee	Large public universities, MoE-wide deployment

7.2 Cost Categories

- **Hosting plus AI plus DB:** Vercel hosting, Z.AI plus ILMU inference, Supabase Postgres and Storage. Scales with submission volume.
- **Staff salary:** engineering, customer success, and product. Two-person team in Year 1; ramps to a small functional team by Year 7.
- **Marketing:** conferences, case-study production, content, and digital outreach. Scales with the sales motion.
- **Legal:** Data Processing Agreements per institution, PDPA compliance work, IP, contracts. Scales with the customer base.

7.3 Revenue Sources

- **Licensing revenue (primary):** annual SaaS licence per active student. Dominant revenue stream across all years.
- **Government grants (secondary):** Cradle Fund, MAGIC, MyDigital, MOHE-administered programmes. Grant intensity drops as commercial revenue grows.
- **Implementation services (Year 3 onwards):** per-procedure SOP authoring assistance and letter-template migration beyond the standard 5 procedures included.
- **ILMU premium routing (Year 5 onwards):** optional surcharge for Malaysia-sovereign LLM routing. Material for government-sector customers.
- **Analytics add-on (Year 7 onwards):** advanced cohort analytics and external reporting integrations (e.g. JPA scholarship reporting). Enterprise tier only.

8. Financial Projections

A 7-year projection, with detail at the four strategic milestone years (Year 1, 3, 5, and 7). Each year has all major revenue and cost lines so the path to profitability is transparent. The operating margin is positive from Year 1 and grows alongside the customer base.

Metric	Y1	Y3	Y5	Y7
Active Institutions	1 (UM pilot)	5	20	40+ (incl. SEA)
Active Students	33K	200K	800K	2M
Licensing Revenue	RM 50K	RM 500K	RM 3M	RM 8M
Government Grants	RM 200K	RM 300K	RM 500K	RM 800K
Hosting + AI + DB	RM 80K	RM 200K	RM 900K	RM 2M
Staff Salary	RM 120K	RM 400K	RM 1.2M	RM 3M
Marketing	RM 10K	RM 50K	RM 150K	RM 400K
Legal	RM 10K	RM 20K	RM 50K	RM 120K
Total Cost	**RM 220K**	**RM 670K**	**RM 2.3M**	**RM 5.5M**
Profit / Loss	**+RM 30K**	**+RM 130K**	**+RM 1.2M**	**+RM 3.3M**

Notes on the projection. Year 1 covers the UM pilot only and is modestly profitable at RM 30K, co-funded by UM Centre of Innovation at a nominal per-seat price; the primary goal is the case study and validation, not scale. Year 3 accelerates with five paying institutions and a growing operating margin. Year 5 marks the start of the ASEAN expansion (Indonesia and Singapore) and a multi-tenant rebuild that absorbs additional hosting plus engineering cost. Year 7 represents a regional ASEAN footprint with 40 plus institutions, RM 8.8M total revenue, and a 37.5% operating margin.

9. Go-to-Market Strategy

Phase 1: UM as Starting Market (Months 0 to 6)

- Co-fund with UM Centre of Innovation and the relevant deans (HEP, IGS). Run on UM's own Supabase and Vercel infrastructure.
- Publish a case study after one academic semester documenting time-to-decision improvements and student-satisfaction lift.

Phase 2: Malaysian Public Sector (Months 6 to 18)

- Use the UM case study to open conversations with USM, UKM, UTM, UPM, and UPSI.
- Formal introduction through MOHE (Ministry of Higher Education); the MY-sovereign-LLM (ILMU) angle is a differentiator versus foreign-stack competitors.
- Lock in 5 customers by Year 3 (the milestone in the financial projection).

Phase 3: Malaysian Private Universities (Months 18 to 36)

- Taylor's, Sunway, Monash MY, Nottingham MY, INTI; different procurement cycles but similar pain points.
- Pricing premium: private universities can absorb the Enterprise tier.

Phase 4: ASEAN Expansion (Year 4 onwards)

- Indonesia first; largest tertiary market in SEA, similar regulatory shape to Malaysia, ASEAN integration tailwinds.
- Singapore for credibility (SMU, NTU, NUS); Thailand and Vietnam to follow.
- Multi-tenant rebuild required (out of MVP scope); see Roadmap.

10. Team & Execution

Team Breaking Bank: a four-person team, all current Universiti Malaya undergraduates.

- **Jeanette Tan En Jie.** Group Leader. Product, business strategy, stakeholder relations.
- **Teo Zen Ben.** Technical Lead. End-to-end engineering, AI architecture, deployment.
- **Nyow An Qi.** Pitch deck design and finals pitch presentation.
- **Thevesh A/L Chandran.** Engineering. Schema, integrations, data quality.

The team's UMHackathon 2026 finals build is shipped, deployed, and measurable. Today's work shipped the parseDocument wiring, SOP-grounded briefing, plan-mode confirmation on Request More Info, coordinator coassist, student AI chat with hybrid escalation, the /admin/cost token-spend dashboard, ten realistic dummy supporting-document PDFs for the demo, and updated non-technical specifications. All schema migrations are applied to production.

11. Risks & Mitigations

Risk	Likelihood × Impact	Mitigation
AI hallucination in decision letters (wrong fact, wrong policy)	Medium × High; directly visible to students	Regex hallucination check; GLM second-layer judge (a separate GLM call cross-verifies every factual claim in the draft letter against the student's record before coordinator review); coordinator-as-final-reviewer; SOP-grounded briefing flags; never auto-send
AI provider outage (Z.AI rate-limit or downtime)	Low × High; would block the loop	Fallback briefing on submit; raw template on letter fill; mock fixtures shipped for demo
PDPA compliance gap	Medium × High; legal liability	RLS on every table; signed URLs; service-role secrets server-only; explicit data-retention policy in DPA

Risk	Likelihood × Impact	Mitigation
Procurement-cycle inertia in public universities	High × Medium; slow revenue ramp	UM reference site; MOHE introduction; co-fund Year 1 with target institution
Cost growth from token consumption at scale	Medium × Medium; gross-margin compression	/admin/cost dashboard for live monitoring; route low-stakes calls to GLM-4.5-flash; cache common SOP retrievals
Foreign-stack chatbot competitor enters the segment	Medium × Medium	ILMU sovereign-MY moat for government sector; deep SOP integration; UM partnership and reference customer

12. Funding & Sustainability

- UM Centre of Innovation co-funding for the pilot deployment plus case-study production.
- Cradle Fund / MAGIC / MyDigital startup grants (dual-track: enterprise SaaS plus GovTech).
- Year-2 seed round (RM 1.5M) once pilot metrics are published, targeting Malaysian VCs and ASEAN-edtech specialists.
- Sustainability path: SaaS revenue alone covers operations from Year 3 onward at the modelled trajectory.

13. Product Roadmap

Detailed in the engineering plan and the non-technical Living Spec. Highlights:

- **Q3 2026:** Email and SMS notification delivery; per-faculty coordinator scoping; OCR for scanned PDFs.
- **Q4 2026:** Tamper-proof audit log (append-only and signed); Bahasa Melayu UI; integration with MAYA, SiswaMail, SPeCTRUM where APIs allow.
- **Q1 2027:** Multi-tenant isolation (foundation for SEA expansion); per-institution branding; native mobile app (PWA first).
- **Q2 2027:** Procedural SDK so universities can author their own procedures without engineering involvement.

Our objective beyond the hackathon is unambiguous: end the silent-fail problem in university paperwork at scale, starting at home.

14. Terms & Definitions

Recurring jargon used throughout this document, defined in plain language.

Term	Plain-language definition
AI hallucination	When the AI states something untrue or invents a fact (for example, a wrong CGPA in a letter). UniGuide's hallucination check is a regex layer that catches these mismatches before any letter is sent.
ARPU	Average Revenue Per User; in our model the average annual revenue per active student under licence.
ASEAN	Association of Southeast Asian Nations; ten member countries including Malaysia, Indonesia, Singapore, Thailand, Vietnam, and the Philippines.
B40 / M40 / T20	The Malaysian Department of Statistics' household-income tiers (B40 = bottom 40%, M40 = middle 40%, T20 = top 20%).
CGPA	Cumulative Grade Point Average; the running academic average on a 4.00 scale.
Coassist	UniGuide's natural-language Revise channel for coordinators: type 'make it softer' or 'add the appeal info' and the AI revises the letter, the proposed step, or the briefing reasoning inline.
EMGS	Education Malaysia Global Services; the agency that processes international-student visas in Malaysia.
Five-minute undo	After a coordinator approves or rejects an application, they have a five-minute window with a live countdown during which they can take the decision back before it commits permanently. The student does not see the letter as final until the five-minute window closes.
FTE	Full-Time Equivalent; one full-time staff member working a full week. '0.2 FTE' means 20% of one full-time staff member.
GLM	Z.AI's family of large language models (GLM-4.6 and GLM-4.5-flash). The mandatory model for UMHackathon 2026.
Happy path / standard case	The straightforward case where nothing unusual happens (e.g. a student with high CGPA and B40 income tier applying for the standard scholarship). Contrast with edge cases needing human discretion.
ILMU	Malaysia's sovereign large language model, developed by YTL AI Labs and Universiti Malaya. Routable as the briefing provider in UniGuide for Bahasa-heavy content.
MOHE	Kementerian Pendidikan Tinggi (Ministry of Higher Education, Malaysia); the regulator overseeing all Malaysian public and private universities.

Term	Plain-language definition
PDPA	Personal Data Protection Act 2010 (Malaysia); the law governing how personal data is collected, processed, and stored.
RLS	Row-Level Security; a database feature that restricts which rows each user can read or write. UniGuide enforces RLS on every table holding personal data.
SaaS	Software as a Service; a subscription model where the customer pays a recurring fee to use software hosted by the vendor.
SLA	Service-Level Agreement; the response-time commitment per submitted case (for example, 'coordinators respond within 48 hours of submit').
SOP	Standard Operating Procedure; the official UM document that describes how a procedure (scholarship, FYP, deferment, etc.) is meant to run. UniGuide's AI reads the SOP at every step.
Triage	In coordinator UX: the inbox tab that surfaces applications a student has flagged for human help, regardless of whether the application has been formally submitted yet.

End of Business Proposal